

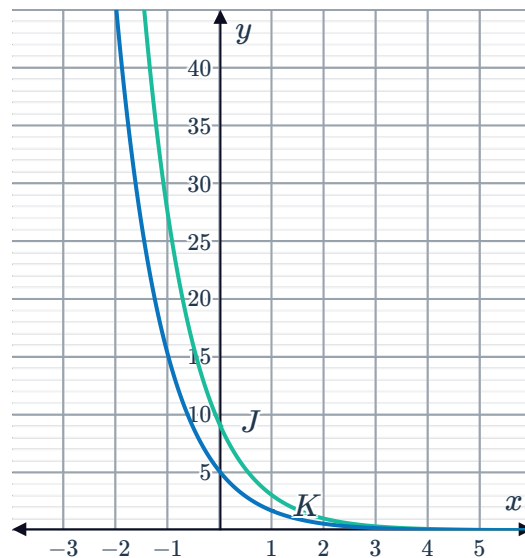
Comparison of exponential functions



1

a Decreasing

b



c As $x \rightarrow \infty$, functions $J, K \rightarrow 0$.

d Dilate $y = 3^{-x}$ vertically by a factor of 5.

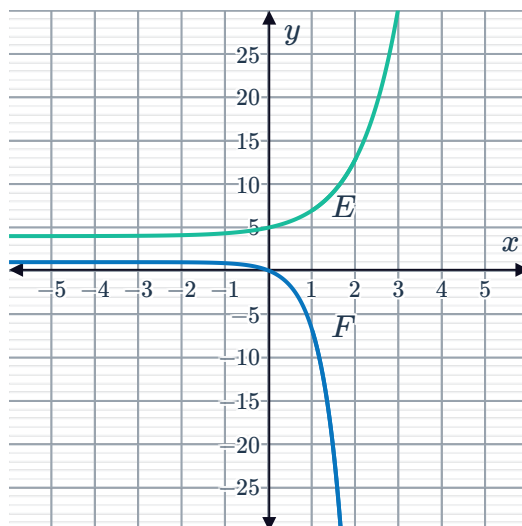
2

a Once

b Function P

3

a



b Zero times

c As $x \rightarrow \infty$, the function $E \rightarrow \infty$ and function $F \rightarrow -\infty$.



Applications

4

a 8 million

b

i 7130 million

ii 9286 million

5

a

i $A = P(1.053)^t$

ii $A = P(1.0125)^{4t}$

b 5.09%

c Bank A

6

a Sale price of property

b 5.3%

c -5.3%

7

a $M = 60\,000(1.032)^t$

b $P = 50\,000(1.061)^t$

c Mohamad's salary will be \$1153.51 greater than Elizabeth's.

d Elizabeth's salary

8

a 30

b 40%

c 100%

d The population of bacteria in the dark environment is increasing at a faster rate than the population of bacteria in the sunlit environment.

9

a $P = 90 + 5t$

b $Q = 90 \times (1.04)^t$

c The current company

d Opening his own business

10

a $P = 100(0.85)^t$

b Fabian

c No, neither Fabian or Zainab will completely forget the content. There is a horizontal asymptote at $y = 0$ and so the recall percentage for both will never reach zero.

- 11 a Yes, there will be a food shortage.



We can derive the equation $F = 3\,374\,000 (1.02)^n$ to represent the growth of the population, and the equation $P = 2\,410\,000 (1.022)^n$ to represent the number of people the food supply will feed.

We can see that the food production is increasing, but at a slower rate than the population's rate of increase, so the population will eventually exceed the amount that can be fed.

- b We can draw the graph of both functions, using technology to find any points of intersection. If the y -value of the population is greater than the y -value of the population that can be fed for the same value of n then there will be a food shortage.

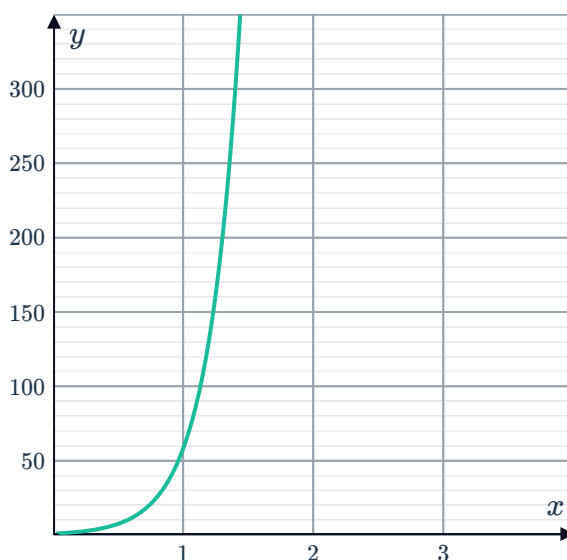
12

- a 179
b 132
c Blair
d 70%
e If you have a high sugar diet, your blood sugar level peaks at greater amounts and decreases more rapidly than a low sugar diet.

13

- a The population of bacteria J increases faster than the population of K .

b



C

